Data Analytics Capstone

Machine-Learning Forecasting of Solar Irradiance and Reliability-Constrained Off-Grid Solar Sizing Across Ghana’s Climatic Zones

Synopsis

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Introduction

Background and Context

Ghana's electricity supply has traditionally relied on hydropower from the Volta basin and thermal power generated from fuel imports, making the grid vulnerable to fluctuations in rainfall and fuel prices. The Government of Ghana's Renewable Energy Master Plan, adopted in 2019, established a target of increasing renewable-energy generation capacity from a 2015 baseline of 42.5 MW to approximately 1,364 MW by 2030, including about 500 MW of utility-scale solar and 200 MW of distributed solar photovoltaics (Energy Commission of Ghana, 2019). Thanks to its tropical climate, Ghana benefits from abundant solar energy, with daily global horizontal irradiance (GHI) of approximately 4-6 kWh/m²/day and a distinct north-south gradient that favors the savannah regions.

Two key groups would benefit immensely from accurate resource forecasting. First, grid operators and developers such as GRIDCo, ECG, and utility-scale developers need short- and medium-term irradiance forecasts for planning generation, reserving capacity, and siting plants. Second, often overlooked in research, ordinary households are increasingly turning to off-grid solar due to unreliable grid power and decreasing PV and battery costs. However, they lack an evidence-based method for sizing installations. Installers often use rules of thumb, such as a fixed number of 'days of autonomy,' that fail to account for regional variations, for example, the high risk of consecutive low-sun days in coastal Accra during the rainy season versus in Navrongo in the Sudan Savanna during Harmattan.

Machine-learning (ML) methods now dominate the international solar-forecasting literature (Voyant et al., 2017; Ahmed et al., 2020), and deep-learning models such as long short-term memory (LSTM) networks have shown 18-34% error reductions over baselines in tropical settings (Qing & Niu, 2018). For Ghana specifically, Quansah et al. (2022) evaluated NASA POWER solar-radiation estimates against ground-station-based estimates derived from sunshine-duration observations at 22 Ghanaian synoptic stations, reporting correlations of approximately r = 0.60-0.94 and supporting NASA POWER as a long-term solar data source for regional comparative analysis. Correlation alone does not establish unbiased agreement; therefore, site-level bias and uncertainty will be examined explicitly during exploratory analysis. The Ghana-specific machine-learning research on solar forecasting identified in the present review is limited and includes plant-level studies such as Asiedu et al. (2024), which examined a 180 kWp grid-connected installation at Tema; the review did not identify published work linking forecasting-grade irradiance modeling to household system sizing.

Problem Statement

Currently, solar resource uncertainty in Ghana is managed using simple averages and heuristics; grid planners lack systematic evidence about which forecasting models are most effective across different climatic zones and seasons. Off-grid households size panels and batteries using fixed heuristics, which can lead to wasted capital through oversizing or blackouts from undersizing. This project addresses two key gaps. First, the literature review conducted for this proposal did not identify a published study that systematically compares classical and machine-learning irradiance forecasts across Ghana's coastal, forest, and savannah zones using a common long-term dataset and formal forecast-comparison tests. Second, the review did not identify a publicly documented, reproducible tool that converts Ghana's historical irradiance record into reliability-constrained, zone-specific off-grid sizing guidance, detailing the number of panels and battery capacity needed to meet a specified consumption level (e.g., 200 kWh/month) at a given reliability level year-round.

The preliminary literature review supporting these gap statements searched Google Scholar, Scopus, ScienceDirect, and IEEE Xplore using combinations of ‘Ghana,’ ‘solar irradiance forecasting,’ ‘solar

PV forecasting,' 'machine learning,' 'NASA POWER,' 'off-grid PV sizing,' 'battery sizing,' and 'loss-of-load probability,' covering publications up to July 2026; the search will be repeated and documented before the final report so that the gap statements remain auditable.

Purpose of the Study

The study has two related objectives. First, it predicts daily GHI by developing and statistically evaluating both traditional (seasonal-naive persistence, SARIMA) and machine learning (random forest, XGBoost, LSTM) models at five locations across Ghana's three climatic zones. This uses twenty (20) years (2005-2024) of NASA POWER daily data and interprets the meteorological factors influencing irradiance using SHAP. Second, it optimizes by transforming the same twenty-year irradiance data into a reliability-constrained off-grid sizing model. This model calculates, for each climatic zone, the PV array size and battery capacity needed to meet a target household load (benchmark: 200 kWh/month) with a specified loss-of-load probability (LOLP). The contribution is practical rather than methodological, emphasizing the multi-zone comparison, seasonal forecast skill stratification (wet/dry/Harmattan), and an end-to-end process linking satellite data to household sizing tables that the present review did not find in prior Ghanaian studies.

Scope and Objectives

The project is bounded to five study sites representing Ghana's climatic zones - Accra (coastal savannah), Takoradi (coastal), Kumasi (moist forest), Tamale (Guinea savannah), and Navrongo (Sudan savannah) - at daily resolution over 2005-2024, with forecast horizons of one day and seven days ahead. Sub-daily forecasting, plant-level power telemetry, and primary data collection are out of scope. Within these bounds, the study pursues five research questions. The forecasting comparison (RQ1-RQ4) is the primary contribution and the sizing application (RQ5) the secondary contribution; this prioritization protects the study if time becomes constrained. Forecast horizons are defined explicitly: the one-day horizon predicts GHI at day t+1, and the seven-day horizon predicts GHI at day t+7, each with a direct forecasting strategy (a separate model trained per horizon), and all predictors are restricted to information available at the forecast origin t.

Research Question 1 (RQ1)

How accurately can machine-learning models forecast daily global horizontal irradiance at representative sites across Ghana's coastal, forest, and savannah climatic zones?

Research Question 2 (RQ2)

Which meteorological variables (temperature, relative humidity, cloud amount, precipitation, wind speed) are most strongly associated with, and most predictive of, daily solar-irradiance variation in Ghana?

Research Question 3 (RQ3)

Do machine-learning models (random forest, XGBoost, LSTM) significantly outperform classical approaches (persistence, day-of-year climatology, and SARIMA-class dynamic harmonic regression) for one-day and seven-day-ahead forecasts?

Research Question 4 (RQ4)

How does forecasting accuracy differ between the wet and dry seasons (including the Harmattan) and across the north-south climatic gradient?

Research Question 5 (RQ5)

What photovoltaic array size and battery capacity does a fully off-grid Ghanaian household require, per climatic zone, to meet a target consumption of 200 kWh per month year-round at a specified reliability level (loss-of-load probability of at most 1-5%)?

The 200 kWh/month benchmark is intended to represent a moderate-to-high household consumption scenario rather than the national median. Energy Commission statistics indicate that Ghana's residential sector consumed on the order of 8,569 GWh in 2024 across 5,281,816 residential customers (Energy Commission of Ghana, 2025), approximately 135 kWh per customer per month on average; this figure will be re-verified against the primary tables of the bulletin during the study, and the benchmark's interpretation is robust to categorization differences because any plausible residential average remains well below 200 kWh/month. Customers are metered connections rather than households, and compound houses commonly share a single meter, so per-household consumption is likely lower than the per-customer average; the consumption distribution is also right-skewed, with a substantial share of customers within the regulator's 0-30 kWh lifeline band. Sizing results are therefore reported for 100, 200, and 300 kWh/month tiers to demonstrate sensitivity to demand level, spanning low-consumption rural households through high-consumption urban households.

Sample Size Calculation

The dataset is a fixed archive rather than a survey to be fielded, so this section assesses whether the planned data volume is adequate for each research question, using confidence-interval (CI) precision and power-of-test methods as appropriate. Because the study analyzes a fixed archive rather than recruiting a sample, and because daily observations are serially dependent, these independent-observation formulas serve only as approximate precision and power benchmarks; final uncertainty estimates and hypothesis tests use time-series-aware procedures (moving-block bootstrap; HLN-corrected Diebold–Mariano tests) that account for the reduction in effective sample size caused by autocorrelation. Throughout, the significance level is α = 0.05 (z = 1.96) and, where power is used, power is 90% (z = 1.28).

RQ1 (CI precision). Estimating mean forecast error with margin of error E = 0.05 kWh/m², taking the standard deviation of daily GHI (conservatively, of daily errors) as σ = 1.0 kWh/m²: $n = \frac{z \times \sigma^2}{E} = \frac{1.96 \times 1.0^2}{0.05} \approx 1{,}537$ daily observations per site.

RQ2 (power, correlation). To detect a weak correlation of ρ = 0.10 at α = 0.05 with 90% power via the Fisher z-transformation, where C(0.10) = 0.1003: $n = \frac{z_{\alpha/2} + z_{\beta}{}^2}{C(\rho)} + 3 = \frac{1.96 + 1.28^2}{0.1003} + 3 \approx 1{,}047$ observations. The archive is fixed, and all available observations will be used; this benchmark confirms adequacy rather than determining collection.

RQ3 (power, paired comparison). Models are compared on the same test days, a paired design on daily error differences. To detect a small, standardized effect d = 0.10: $n = \frac{z_{\alpha/2} + z_{\beta}{}^2}{d} = \frac{1.96 + 1.28^2}{0.10} \approx 1{,}051$ paired test days.

RQ4 (power, two-sample). Comparing wet-season versus dry-season forecast errors with effect size d = 0.20: $n = 2\frac{z_{\alpha/2} + z_{\beta}{}^2}{d} = 2\frac{1.96 + 1.28^2}{0.20} \approx 526$ days per season group.

RQ5 (CI precision, proportion). LOLP is a proportion (the share of observed historical days on which the modeled system would fail to serve the load). To estimate an LOLP near p = 0.01 within E = 0.005 at 95% confidence: $n = \frac{z^2\, p(1-p)}{E^2} = \frac{3.8416 \times 0.0099}{0.000025} \approx 1{,}522$ historical days evaluated per zone-configuration. Loss-of-load days occur in runs rather than independently, so the effective sample size is smaller than the calendar-day count; interval estimates for LOLP therefore use year-block and moving-block bootstrap methods.

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Confidence interval (mean)	$z = 1.96$, $\sigma = 1.0$ kWh/m², $E = 0.05$	1,537 days/site
RQ2	Power of test (correlation, Fisher z)	$\alpha = .05$, power = .90, $\rho = 0.10$	1,047 days
RQ3	Power of test (paired difference)	$\alpha = .05$, power = .90, $d = 0.10$	1,051 test days
RQ4	Power of test (two-sample)	$\alpha = .05$, power = .90, $d = 0.20$	526 days/season
RQ5	Confidence interval (proportion)	$z = 1.96$, $p = 0.01$, $E = 0.005$	1,522 days/zone

Table 1. Minimum sample size per research question.

Final sample size: N = max(RQ1-RQ5) = 1,537 daily observations per site (about 4.2 years of daily data). The project retrieves twenty years (2005-2024) of daily data - approximately 7,305 observations per site and 36,525 in total across the five sites - exceeding the binding requirement by a factor of more than four. For RQ3, the held-out test period provides about 730 paired days per site, which is below the 1,051-day benchmark for a single site; the requirement is met by pooling test days across the five sites (about 3,650 paired observations), with per-site comparisons reported descriptively. The RQ4 per-season requirement is likewise met by pooling across sites (about 1,825 test days per season group against the 526 required), and the full twenty-year record supports the RQ5 historical backtest at all five sites simultaneously.

Data Description

The primary data source is the NASA Prediction of Worldwide Energy Resources (POWER) archive (NASA Langley Research Center, 2024), which provides daily satellite-derived solar and meteorological parameters on a global grid, free of charge and without registration, through a documented REST API.

Quansah et al. (2022) validated POWER solar-radiation products against ground-based estimates at 22 Ghanaian synoptic stations (r = 0.60-0.94), with strongest agreement in the northern savannah, supporting its use for this study. Data are retrieved for the five study sites listed in Table 2.

Data source URLs:

- Archive and documentation: https://power.larc.nasa.gov/

- Example API call (Accra, all six parameters, 2005-2024):

https://power.larc.nasa.gov/api/temporal/daily/point?parameters=ALLSKY_SFC_SW_DWN,T2M,RH2M,CLOUD_AMT,PRECTOTCORR,WS2M&start=20050101&end=20241231&latitude=5.60&longitude=-0.19&community=RE&format=CSV

Site	**Climatic zone**	**Latitude**	**Longitude**
Accra	Coastal savannah	5.60 N	0.19 W
Takoradi	Coastal	4.90 N	1.76 W
Kumasi	Moist semi-deciduous forest	6.69 N	1.62 W
Tamale	Guinea savannah	9.40 N	0.84 W
Navrongo	Sudan savannah	10.89 N	1.09 W

Table 2. Study sites spanning Ghana's three climatic zones.

All download scripts, raw and processed data, notebooks, and results will be maintained in a public GitHub repository (URL: https://github.com/kudumens/capstone-solar-ghana) with the following structure; the accompanying public GitHub repository already implements the download, validation, and cleaning pipeline.

```
capstone-solar-ghana/
|-- README.md               (reproduction instructions)
|-- data/raw/               (NASA POWER CSVs, one per site)
|-- data/processed/         (cleaned, feature-engineered datasets)
|-- notebooks/              (01_download ... 07_offgrid_sizing)
|-- src/                    (reusable Python modules)
|-- results/                (metrics tables, figures, sizing tables)
```

```
`-- reports/                (interim and final report assets)
```

Data Dictionary (Mandatory)

Variable (POWER code)	Meaning	Type / Units	Role
ALLSKY_SFC_SW_DWN	All-sky global horizontal irradiance	Continuous, kWh/m²/day	Dependent (Y)
T2M	Air temperature at 2 m	Continuous, °C	Predictor
RH2M	Relative humidity at 2 m	Continuous, %	Predictor
CLOUD_AMT	Cloud amount	Continuous, %	Predictor
PRECTOTCORR	Corrected total precipitation	Continuous, mm/day	Predictor
WS2M	Wind speed at 2 m	Continuous, m/s	Predictor
Day-of-year / month	Calendar position (cyclically encoded)	Cyclical numeric	Engineered predictor
Season	Wet / dry / Harmattan indicator	Categorical	Predictor / stratifier
Lagged GHI (t-1 ... t-7)	Recent irradiance history	Continuous, kWh/m²/day	Engineered predictor
Rolling means (7-, 30-day)	Smoothed irradiance trend	Continuous, kWh/m²/day	Engineered predictor
Site / zone	Location identifier	Categorical	Grouping variable
Modeled battery state-of-charge (derived from observed data)	Daily energy balance in RQ5 historical backtest	Continuous, kWh	RQ5 state variable

Table 3. Data dictionary.

Analytic Approach

Each research question is paired with a testable hypothesis and mapped to statistical and machine-learning techniques. The overall pipeline is API ingestion → cleaning and validation → feature engineering (lags, rolling means, cyclical calendar encoding, season indicators) → strict temporal split (train 2005-2020, validate 2021-2022, test 2023-2024; no shuffling) → baseline models → ML models

→ statistical comparison → SHAP interpretation → off-grid sizing backtest. Tools are Python 3 (pandas, statsmodels, scikit-learn, XGBoost (Chen & Guestrin, 2016), TensorFlow/Keras, SHAP (Lundberg & Lee, 2017)) with Git/GitHub for reproducibility; all are free and open-source. All preprocessing parameters (scaling, imputation) will be estimated on training data only, and lagged and rolling predictors will be constructed from strictly prior observations to minimize temporal leakage; hyperparameters will be selected using expanding-window (rolling-origin) time-series cross-validation within the training and validation periods (Hyndman & Athanasopoulos, 2021). No synthetic or artificially generated data are used anywhere in the study: every analysis, including the RQ5 off-grid sizing, operates exclusively on the observed NASA POWER record. RQ5 is a historical backtest in which a deterministic energy-balance calculation is applied to each actually observed day; no weather or irradiance values are synthesized, resampled, or randomly generated, and the only assumed quantities are fixed engineering constants (performance ratio, battery depth-of-discharge) taken from the published literature.

Predictor availability and look-ahead bias. To prevent look-ahead bias, all predictors used for an h-day-ahead forecast are restricted to values available at or before the forecast origin t: lagged GHI, lagged meteorological variables, and deterministic calendar features. Contemporaneous realized meteorological values for the target day are not used as forecast inputs, because they would not be available at operational forecast issuance; same-day meteorological variables enter only the separate explanatory analysis for RQ2. Formally, the information set is

$$GH\hat{I}_{t+h} \;=\; f(GHI_t, GHI_{t-1}, \dots, X_t, X_{t-1}, \dots, C_{t+h}),\;\; h \in \{1, 7\}$$

where X denotes lagged meteorological predictors, and C denotes deterministic calendar features (day-of-year harmonics and season indicators), which are known in advance for any future date.

Null and alternative hypotheses. Each research question is validated by an explicit hypothesis test conducted at significance level $\alpha = 0.05$:

RQ1 — H_0: Skill ≤ 0 (the model does not improve on the classical benchmark); H_a: Skill > 0, where Skill = 1 − RMSE_model / RMSE_benchmark. Assessed with moving-block-bootstrap confidence intervals on the untouched 2023–2024 test window (Künsch, 1989). RQ1 is otherwise primarily evaluative: RMSE, MAE, normalized RMSE, and R^2 are reported descriptively with confidence intervals, and $R^2 \geq 0.80$ is retained as a descriptive performance target rather than a test criterion, since irradiance predictability varies with horizon, location, and season.

RQ2 — for each meteorological driver j: H_0: $\rho_j = 0$ (no association between driver j and daily GHI); H_a: $\rho_j \neq 0$. Tested with Pearson and Spearman correlation tests, applying the Holm correction for multiple comparisons across the five drivers; relative predictor strength is then ranked with permutation importance and SHAP values, which are interpreted as model-specific predictive contributions rather than causal effects (Lundberg & Lee, 2017).

RQ3 — H_0: $E[d_t] = 0$, where d_t is the daily difference in squared forecast errors between the ML model and the classical baseline (equal predictive accuracy); H_a: $E[d_t] < 0$ (the ML model is more accurate). Tested with the one-sided Diebold–Mariano test (Diebold & Mariano, 1995) using the Harvey–Leybourne–Newbold small-sample correction (Harvey et al., 1997); a moving-block-bootstrap confidence interval for the mean loss differential (Künsch, 1989) serves as a dependence-robust check. Because comparisons span multiple models, sites, and horizons, p-values within the model-comparison family are adjusted with the Holm procedure.

RQ4 — for the mean comparison, H_0: E(|e|_wet) = E(|e|_dry) versus H_a: the mean absolute errors differ (two-sample t-test); for the distributional comparison, H_0: F_wet(e) = F_dry(e) versus H_a: the seasonal error distributions differ (Mann–Whitney U). Across zones, H_0: error distributions are equal in all five zones; H_a: at least one zone differs; tested with one-way ANOVA or Kruskal–Wallis. Season boundaries are defined per zone through an explicit month mapping reflecting Ghana's bimodal southern and unimodal northern rainfall regimes, so wet, dry, and Harmattan periods are not assumed identical across sites.

RQ5 — H_0: the minimum battery capacity required to meet a fixed LOLP target is equal across the five climatic zones (B_Accra = B_Takoradi = B_Kumasi = B_Tamale = B_Navrongo); H_a: at least one zone's requirement differs. Assessed by comparing zone-level capacity requirements with block-bootstrap confidence intervals computed from the twenty-year observed record.

These tests correspond to the following directional research expectations. H1 (for RQ1): Machine-learning models achieve R^2 of at least 0.80 for one-day-ahead GHI forecasts in each climatic zone. H2 (for RQ2): Cloud amount and relative humidity have significantly stronger associations with daily GHI than temperature, precipitation, or wind speed. H3 (for RQ3): ML models reduce RMSE by at least 15% relative to the seasonal-naive baseline, and the paired difference in daily errors is significant at $\alpha = 0.05$. H4 (for RQ4): Forecast RMSE differs significantly between wet and dry seasons, with the largest degradation in the southern zones during June-August. H5 (for RQ5): Required battery capacity for a fixed reliability target differs across climatic zones by at least 25%, implying that uniform national sizing rules materially mis-size systems.

Solution to RQ1-RQ5

RQ1 (statistical + ML): Train random forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016), and LSTM (Hochreiter & Schmidhuber, 1997) models per site, with a pooled multi-site LSTM also evaluated for data efficiency; evaluate on the untouched 2023-2024 test window with RMSE, MAE, normalized RMSE, and R^2, reporting 95% confidence intervals obtained by moving-block bootstrap over test days (Künsch, 1989). LSTM specification: 30-day input windows; one to two recurrent layers of 32-64 units; dropout; early stopping on the validation window; Adam optimizer; and five repeated runs with fixed random seeds to control stochastic training variation.

RQ2 (statistical + ML): Pearson and Spearman correlations with multicollinearity screening (variance inflation factors), complemented by permutation feature importance and SHAP values on the tree models; hypothesis tests on correlation coefficients at $\alpha = 0.05$.

RQ3 (statistical): Paired same-day comparison of model errors against the classical benchmarks — persistence (the forecast equals the last observed value), day-of-year climatology (the forecast equals the historical mean GHI for that calendar day), and a SARIMA-class model implemented as dynamic harmonic regression (Fourier seasonal terms with ARIMA errors), which handles the ≈365-day annual cycle more efficiently than a large seasonal SARIMA structure (Hyndman & Athanasopoulos, 2021) — using the HLN-corrected Diebold–Mariano test with Holm adjustment across the comparison family, with the skill score defined below as the effect-size measure.

RQ4 (statistical): Stratify test-set errors by season (wet, dry, Harmattan) and zone; compare distributions with two-sample t-tests or Mann-Whitney U where normality fails; report seasonal skill profiles per zone.

RQ5 (historical backtest + optimization): Drive a daily energy-balance backtest of a PV-plus-battery system through the full twenty-year observed irradiance record per zone. Daily PV energy is

$$E_{PV,t} = P_{PV} \times H_t \times PR$$

where H_t is the observed daily GHI and PR is the system performance ratio, set to 0.75 with inverter and wiring losses folded in, within the ranges reviewed by Khatib et al. (2013). The battery is modeled with explicit efficiencies and state-of-charge (SOC) limits:

$$SOC_t = min(C_{max}, \quad SOC_{t-1} + \eta_c E_{charge,t} - \frac{E_{discharge,t}}{\eta_d})$$

$$SOC_{min} \le SOC_t \le C_{max}, \; SOC_{min} = (1 - DoD_{max})\, C_{max}$$

with charge and discharge efficiencies ηc = ηd = 0.95 (approximately 90% round-trip), maximum depth-of-discharge DoDmax = 0.80 (Khatib et al., 2013), surplus generation curtailed once the battery is full, and unmet energy recorded whenever the load exceeds available supply. The first year of the record (2005) serves as a warm-up period to remove dependence on the assumed initial state of charge and is excluded from reliability scoring. The household load is applied as a flat daily profile (200 kWh/month = 6.67 kWh/day), with sensitivity analysis for moderate day-to-day and seasonal load variability. For each candidate configuration (array kWp, battery kWh), reliability is scored with the three complementary metrics defined below (LOLP, LPSP, EENS), and a grid search identifies the smallest feasible PV–battery configurations meeting each reliability target (LOLP ≤ 1% and ≤ 5%), reported as a PV-versus-battery trade-off frontier rather than a single minimum-cost design, because component prices are outside the study's scope. Because the backtest runs at daily resolution, it captures multi-day resource variability but not the intraday timing of generation and demand; the resulting capacities are therefore planning estimates rather than installation-ready engineering designs. Forecast models from RQ1 additionally provide an operational layer: day-ahead low-sun warnings that allow households to shift discretionary loads.

Evaluation Metrics (Formulae and Calculations)

Point-forecast accuracy is evaluated with the root-mean-square error, mean absolute error, coefficient of determination, and forecast skill relative to the classical baseline:

$$RMSE = \sqrt{\frac{1}{n} \Sigma_t (y_t - \hat{y}_t)^2}$$

$$MAE = \frac{1}{n} \Sigma_t | y_t - \hat{y}_t |$$

$$R^2 = 1 - \frac{\Sigma_t (y_t - \hat{y}_t)^2}{\Sigma_t (y_t - \bar{y})^2}$$

$$Skill = 1 - \frac{RMSE_{model}}{RMSE_{baseline}}$$

System reliability in RQ5 is scored with three complementary metrics: the loss-of-load probability (frequency of failure days), the loss-of-power-supply probability (share of total energy demand unserved), and the expected energy not served (magnitude of the shortfall). N_unmet is the number of observed historical days on which the load could not have been served; a system with one small deficit and one with a full-day outage are thereby distinguished by LPSP and EENS even when their LOLP is identical:

$$LOLP = \frac{N_{unmet}}{N_{total}}$$

$$LPSP = \frac{\Sigma_t E_{unmet,t}}{\Sigma_t E_{load,t}}$$

$$EENS = \Sigma_t E_{unmet,t}$$

Worked calculation (success criterion). Success on RQ3 requires Skill >= 0.15. Illustration: if the seasonal-naive baseline achieves RMSE = 0.80 kWh/m^2 on the test window and XGBoost achieves RMSE = 0.64 kWh/m^2, then Skill = 1 - 0.64/0.80 = 0.20 >= 0.15, satisfying the criterion; the Diebold-Mariano test then determines whether the improvement is statistically significant. Worked calculation (RQ5 benchmark). A 200 kWh/month household draws 6.67 kWh/day. Sizing on the worst-month resource rather than the annual mean: with worst-month peak-sun-hours of 4.0 (southern zones, June-August) and performance ratio PR = 0.75, required array = $6.67 / (4.0 \times 0.75) \approx 2.2$ kWp, rounded upward to six 400 W modules (2.4 kWp); a two-day-autonomy battery at 80% usable depth-of-discharge is $2 \times 6.67 / 0.8 \approx 16.7$ kWh. The backtest may identify a different PV–battery trade-off than these deterministic starting values. These deterministic figures are the starting grid for the LOLP search, which replaces the assumed two-day autonomy with the empirically observed worst runs of consecutive low-sun days in each zone's twenty-year record.

Recommendation and Application

Target users. (1) Off-grid and grid-defecting households and the solar installers who serve them: the study's headline deliverable is a zone-by-zone sizing table - panels and battery kWh required in Accra, Kumasi, Takoradi, Tamale, and Navrongo for standard consumption tiers (100, 200, 300 kWh/month) at 99% and 95% reliability - replacing one-size-fits-all autonomy rules with evidence. A household in Navrongo should not pay for the same battery bank as one in Takoradi, and the study will quantify exactly how different their needs are. (2) Grid operators (GRIDCo, ECG): day-ahead and week-ahead irradiance forecasts with quantified zone- and season-specific skill, informing dispatch and reserve planning as solar penetration grows toward the Renewable Energy Master Plan targets. (3) Developers and financiers: zone-level resource reliability and forecastability profiles for siting and bankability assessment. (4) Policy makers (Energy Commission): evidence on where solar output is most predictable and where off-grid economics are strongest, informing siting incentives and rural electrification strategy. All outputs are released as a reproducible open pipeline that can be re-run for any West African location covered by NASA POWER.

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